

Airline Ontology Analytics

Hersteller-Segmente & On-Time Performance (OTP)
2000 – 2008

Datenquelle: `airlinedata.flights_orc` + `planes`

Ontologie: `Flight` → `assignedAircraft` → `Plane.manufacturer`

OTP: FAA 15-min-Regel (`OnTimeFlight` / abgeschlossene Flüge)

Erstellt: 2026-09-11

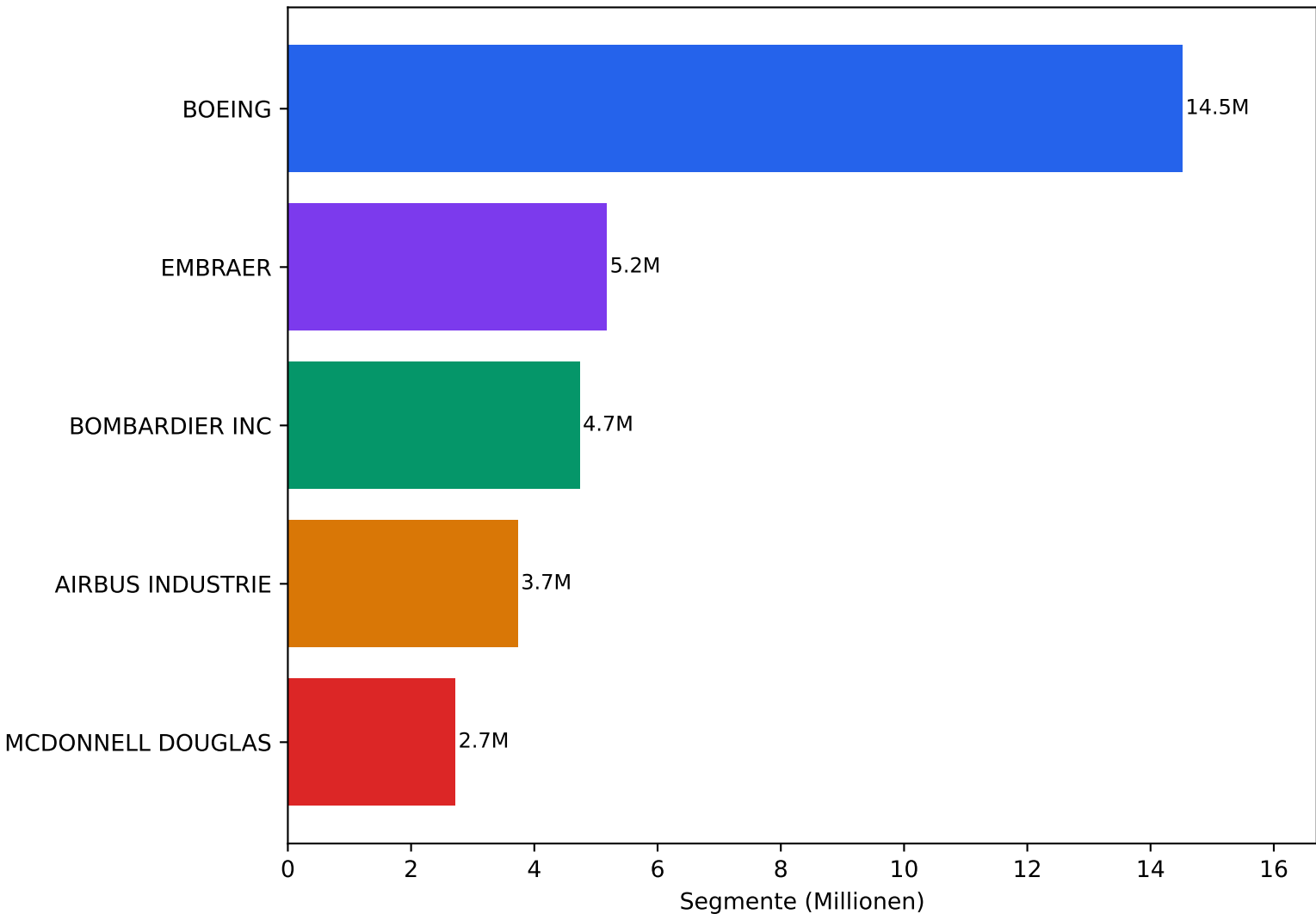
Zusammenfassung

- Boeing dominiert mit ~14,5 Mio. Segmenten ($\approx 48\%$ der Top-5).
- EMBRAER und Bombardier folgen mit ~5,2 Mio. bzw. ~4,7 Mio. Segmenten.
- OTP sinkt branchenweit von ~80-84 % (2002-2003) auf ~70-76 % (2006-2008).
- 2007 war das schwächste OTP-Jahr bei Airbus, Bombardier und McDonnell Douglas.
- Embraer/Bombardier: 2000-2002 nur wenige tailnum-Matches — OTP dort unsicher.

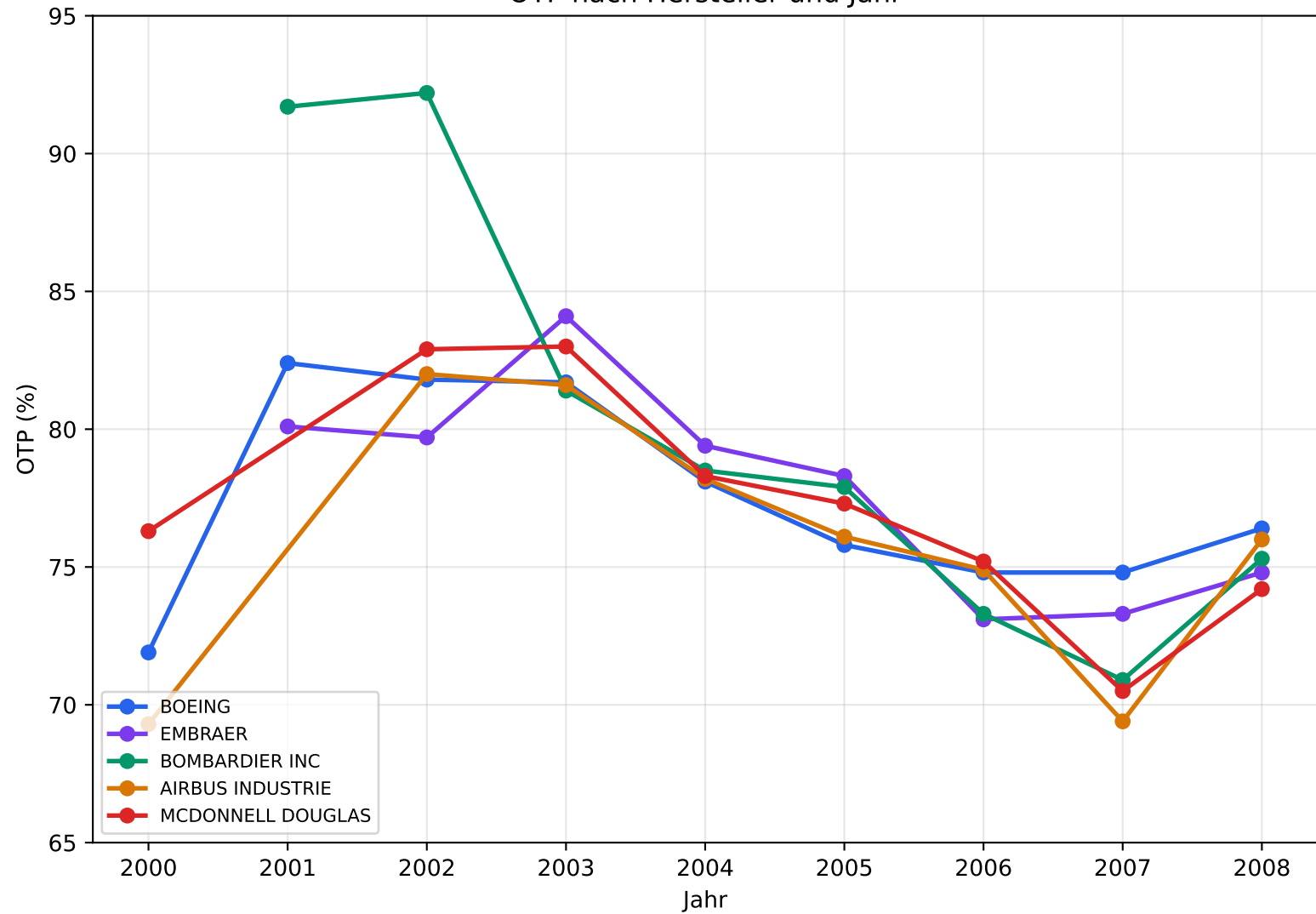
Top 5 Hersteller — Segmente gesamt (2000-2008)

Hersteller	Segmente
BOEING	14.518.058
EMBRAER	5.176.326
BOMBARDIER INC	4.735.636
AIRBUS INDUSTRIE	3.733.463
MCDONNELL DOUGLAS	2.717.289

Flugsegmente nach Hersteller (2000-2008)



OTP nach Hersteller und Jahr



BOEING — Segmente & OTP pro Jahr

Jahr	Segmente	OTP %
2000	1.198.090	71.9
2001	250.612	82.4
2002	1.067.163	81.8
2003	1.291.371	81.7
2004	1.710.370	78.1
2005	1.792.098	75.8
2006	1.866.999	74.8
2007	2.679.778	74.8
2008	2.661.577	76.4

EMBRAER — Segmente & OTP pro Jahr

Jahr	Segmente	OTP %
2001	281	80.1
2002	75	79.7
2003	480.102	84.1
2004	714.051	79.4
2005	973.462	78.3
2006	1.057.262	73.1
2007	1.063.685	73.3
2008	887.408	74.8

BOMBARDIER INC — Segmente & OTP pro Jahr

Jahr	Segmente	OTP %
2001	1.916	91.7
2002	357	92.2
2003	359.764	81.4
2004	670.367	78.5
2005	791.576	77.9
2006	930.137	73.3
2007	1.006.829	70.9
2008	974.690	75.3

AIRBUS INDUSTRIE — Segmente & OTP pro Jahr

Jahr	Segmente	OTP %
2000	309.941	69.3
2002	386.936	82.0
2003	462.052	81.6
2004	470.468	78.2
2005	495.007	76.1
2006	511.871	74.9
2007	559.454	69.4
2008	537.734	76.0

MCDONNELL DOUGLAS — Segmente & OTP pro Jahr

Jahr	Segmente	OTP %
2000	361.883	76.3
2002	291.254	82.9
2003	341.000	83.0
2004	350.409	78.3
2005	347.017	77.3
2006	344.144	75.2
2007	352.229	70.5
2008	329.353	74.2

Semantica Reasoning — Übersicht

Semantica Reasoner — Forward & Backward Chaining

Narrative:

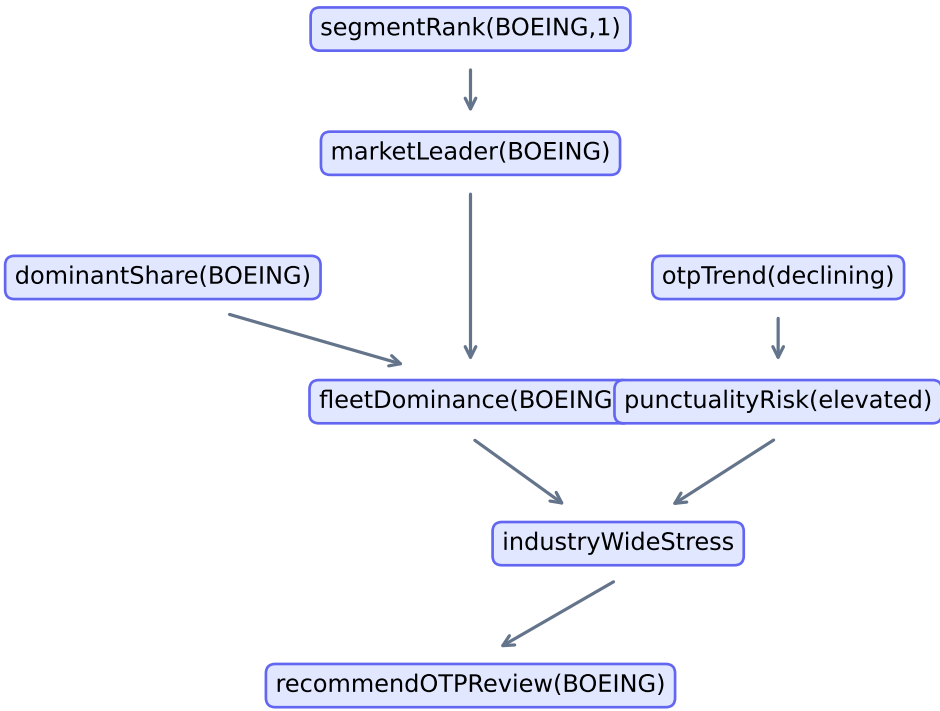
1. Ontologie-Auflösung: Flight → assignedAircraft → Plane.manufacturer auf flights_orc + planes.
2. Aggregation 2000–2008: Boeing führt mit 14,5 Mio. Segmenten (segmentRank=1 → marketLeader).
3. OTP-Klassifikation: FAA 15-min-Regel (OnTimeFlight) pro Jahr und Hersteller.
4. Trend 2003→2007: otpTrend(declining) bei allen Top-5 → punctualityRisk(elevated).
5. Schwachjahr 2007: weakOtp2007 für Airbus, Bombardier, McDonnell Douglas.
6. Boeing >40% Top-5-Anteil → fleetDominance(BOEING).
7. Marktführer + sinkende OTP → industryWideStress(2000-2008).
8. Empfehlung: recommendOTPReview(BOEING) — dominante Flotte, fallende Pünktlichkeit.

Decision ID (airline_graph.json): dbc6a178-b5f0-4cef-b29c-6283965b8b57

Haupt-Empfehlung: recommendOTPReview(BOEING)

Begründung: fleetDominance + punctualityRisk(elevated)

Reasoning-Kette (Kernpfad)



Reasoning-Regeln

Forward-chaining Regeln (config/airline_manufacturer_reasoning.yaml):

1. IF segmentRank(?m, 1) THEN marketLeader(?m)
2. IF marketLeader(?m) AND dominantShare(?m) THEN fleetDominance(?m)
3. IF otpTrend(?m, declining) THEN punctualityRisk(?m, elevated)
4. IF weakOtp2007(?m) THEN worstYearCandidate(?m, 2007)
5. IF punctualityRisk(?m, elevated) AND marketLeader(?m) THEN industryWideStress(2000-2008)
6. IF fleetDominance(?m) AND punctualityRisk(?m, elevated) THEN recommendOTPReview(?m)
7. IF sparseTailnumMatch(?m) THEN dataQualityCaveat(?m)

Inference Trace

Abgeleitete Fakten (mit Premissen):

→ marketLeader(BOEING)

Regel: rule_1 | Premissen: segmentRank(BOEING, 1)

→ fleetDominance(BOEING)

Regel: rule_2 | Premissen: marketLeader(BOEING), dominantShare(BOEING)

→ punctualityRisk(AIRBUS_INDUSTRIE, elevated)

Regel: rule_3 | Premissen: otpTrend(AIRBUS_INDUSTRIE, declining)

→ punctualityRisk(BOEING, elevated)

Regel: rule_3 | Premissen: otpTrend(BOEING, declining)

→ punctualityRisk(BOMBARDIER_INC, elevated)

Regel: rule_3 | Premissen: otpTrend(BOMBARDIER_INC, declining)

→ punctualityRisk(EMBRAER, elevated)

Regel: rule_3 | Premissen: otpTrend(EMBRAER, declining)

→ punctualityRisk(MCDONNELL_DOUGLAS, elevated)

Regel: rule_3 | Premissen: otpTrend(MCDONNELL_DOUGLAS, declining)

→ worstYearCandidate(AIRBUS_INDUSTRIE, 2007)

Regel: rule_4 | Premissen: weakOtp2007(AIRBUS_INDUSTRIE)

→ worstYearCandidate(BOMBARDIER_INC, 2007)

Regel: rule_4 | Premissen: weakOtp2007(BOMBARDIER_INC)

→ worstYearCandidate(MCDONNELL_DOUGLAS, 2007)

Inference Trace (cont.)

Regel: rule_4 | Premissen: weakOtp2007(MCDONNELL_DOUGLAS)

→ industryWideStress(2000-2008)

Regel: rule_5 | Premissen: punctualityRisk(BOEING, elevated), marketLeader(BOEING)

→ recommendOTPReview(BOEING)

Regel: rule_6 | Premissen: fleetDominance(BOEING), punctualityRisk(BOEING, elevated)

→ dataQualityCaveat(BOMBARDIER_INC)

Regel: rule_7 | Premissen: sparseTailnumMatch(BOMBARDIER_INC)

→ dataQualityCaveat(EMBRAER)

Regel: rule_7 | Premissen: sparseTailnumMatch(EMBRAER)

Zielbeweise

Backward-chaining — bewiesene Ziele:

marketLeader(BOEING) — ✓ bewiesen

 Premissen: marketLeader(BOEING)

recommendOTPReview(BOEING) — ✓ bewiesen

 Premissen: recommendOTPReview(BOEING)

industryWideStress(2000-2008) — ✓ bewiesen

 Premissen: industryWideStress(2000-2008)